High-Performance Solid-State Supercapacitors and Microsupercapacitors Derived from Printable Graphene Inks

*Lei Li, Ethan B. Secor, Kan-Sheng Chen, Jian Zhu, Xiaolong Liu, Theodore Z. Gao, Jung-Woo T. Seo, Yichao Zhao, and Mark C. Hersam**

Advances in thin-film energy storage technologies are required to power the emerging field of printed and portable electronics, with applications spanning biomedical and environmental monitoring, flexible displays and logic, and wireless identification and communication to enable the Internet of Things.[1,2] Supercapacitors (SCs), also known as electrochemical capacitors, represent a promising technology to meet this onboard energy storage requirement. SCs store energy in electrochemical double layers formed by highly reversible ion adsorption processes at the interface between electrodes and electrolyte.[3–6] More importantly, the development of microsupercapacitors (MSCs) is facilitating the growth of modern microelectronic systems.[4] All-solid-state SCs and MSCs are particularly well-suited for portable energy storage due to their excellent power density, suitable energy density, long operational lifespan, straightforward packaging with no liquid components, and versatile form factor compatible with thin-film electronic systems.[5,7,8] Since the electrode has a critical impact on the energy storage performance of SCs, significant effort has been devoted to the development and evaluation of electrode materials.[3] Among these materials, graphene is a leading candidate due to its high intrinsic electrical conductivity, resilient mechanical properties, high theoretical surface area, and promising theoretical capacitance.[8–10]

Several approaches have been developed to prepare graphene-based electrodes for SCs and MSCs. Chemical methods have been widely used to synthesize chemically converted graphene,[11] which entail the synthesis of graphene oxide (GO) by the Hummers method or related methods,[12–14] followed by reduction with hydrazine, $NaBH_4$, H_2SO_4, hydrothermal reaction, thermal treatment in inert gas, or laser irradiation.[7,15–21] However, the application of chemically converted graphene in SCs and MSCs has been hindered by several issues. In particular, the synthesis, postreaction treatment, and instability of GO present processing challenges for widespread application.[14,22] In addition, the complex fabrication required for GO-based electrodes limits the cost potential and versatility of devices, particularly for MSCs with interdigitated structures.[4,23] Chemical vapor deposition has also been used to prepare graphene directly, but has limited scalability and often requires harsh synthetic conditions.[24–29] Laser scribing was recently developed to prepare porous graphene networks from polyimide.[30,31] This technology simplifies the fabrication processes of MSCs compared to the photolithography approach, but still suffers from several limitations, such as limited compatibility with other substrates beyond polyimide and low energy storage ability.[4]

Inkjet printing offers a desirable suite of advantages, including additive, non-contact, digital fabrication with high spatial resolution. Moreover, inkjet printing is an industrially mature technology, facilitating rapid transfer of technology from prototyping to manufacturing. The process and substrate compatibility achieved by this method, along with its commercial relevance, make the fabrication of microsupercapacitors by inkjet printing highly advantageous. However, challenges remain in the development of suitable graphene-based inks combining process compatibility and desirable electrical performance. Consequently, the development of a facile and scalable method for the fabrication of graphene electrodes for high-performance SCs and MSCs remains an outstanding challenge.

Recent reports have demonstrated liquid-phase exfoliation of graphite for the production of stable graphene dispersions using the polymer ethyl cellulose in common, low-cost solvents such as ethanol and terpineol.[32–35] The graphene/ethyl cellulose (G/EC) system is suitable for applications in scalable flexible electronics, with demonstrated processing ease and compatibility with a range of desirable substrates, as well as excellent electrical conductivity and mechanical flexibility. Moreover, this system can be tailored for a range of additive manufacturing technologies including inkjet, gravure, and screen printing.[32,36] Herein, we extend this promising processing platform to electrochemical energy storage applications, realizing high-performance solid-state SCs. The suitability of the G/EC material for all-solid-state SC applications is first evaluated using blade-coated and spin-coated thin-film electrodes in sandwich-structured devices. In this configuration, the high-conductivity, binder-free electrode mitigates the need for a separate current collector, simplifying the device fabrication process and eliminating potentially weak interfaces to enable

Dr. L. Li, E. B. Secor, Dr. K.-S. Chen, Dr. J. Zhu, T. Z. Gao, Dr. J.-W. T. Seo, Y. Zhao, Prof. M. C. Hersam
Department of Materials Science and Engineering
Northwestern University
2220 Campus Drive, Evanston, IL 60208, USA
E-mail: m-hersam@northwestern.edu

X. Liu
Graduate Program in Applied Physics
Northwestern University
Evanston, IL 60208, USA

Prof. M. C. Hersam
Department of Chemistry
Department of Electrical Engineering and Computer Science
Northwestern University
Evanston, IL 60208, USA

DOI: 10.1002/aenm.201600909

 © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

mechanically flexible and durable devices. The established compatibility of G/EC with drop-on-demand inkjet printing is then leveraged to fabricate MSCs, demonstrating the versatile utility of printed graphene electrodes for robust, high-performance energy storage applications. Moreover, the solution-processed G/EC ink shows desirable process compatibility with other substrates beyond polyimide, and direct integration with other printed electronic systems.[35,37]

Sandwich-structured SCs were prepared with graphene electrodes as illustrated in Figure S1 (Supporting Information). G/EC is first prepared by liquid-phase exfoliation of graphite in ethanol as described previously.[32,35] Briefly, graphite was exfoliated by high shear mixing to produce few-layer graphene flakes with a typical thickness of ≈2 nm, which were stabilized at high concentration by the polymer EC. Centrifugation and selective flocculation steps were performed to remove remaining graphite flakes and excess EC, yielding a powder of G/EC. In this manner, the ink formulation for coating or printing is largely decoupled from the original exfoliation, providing a versatile platform to tune ink properties. The resulting G/EC powder was dispersed in a mixture of ethanol and terpineol to form an ink that was cast into thin films by spin coating or blade coating onto flexible polyimide substrates, followed by thermal annealing to decompose the EC in the film. The thickness of the graphene film was controlled by adjusting the spin rate or ink concentration, and is designated as G-*X* (where *X* represents the thickness of the film in nanometers). The gel electrolyte of poly(vinyl alcohol)-phosphoric acid (PVA-H_3PO_4) was subsequently deposited on the film surface. After solidification of the electrolyte, two identical electrodes were sandwiched to form the device, identified as SC-G-*X*, as shown in the digital photograph in Figure S1 (Supporting Information). The total thickness of the gel electrolyte in the devices is ≈15 µm (Figure S2, Supporting Information). Details for G/EC synthesis and device fabrication can be found in the Experimental Section.

The morphology of the graphene films was studied by scanning electron microscopy (SEM) and transmission electron microscopy (TEM). **Figure 1**a,b shows top-view SEM images of graphene films at different magnifications, in which the graphene flakes are uniformly packed. In addition, Figure 1c and Figure S3 (Supporting Information) show cross-sectional SEM images of coated graphene films. The average thickness of these films depends on the processing parameters, and ranges from 30 to 2000 nm. Devices assembled using these films are referred to as SC-G-30, SC-G-40, SC-G-50, SC-G-75, SC-G-100, and SC-G-2000, corresponding to the different thicknesses (in nanometers). The morphologies of individual graphene flakes were further characterized by TEM, as shown in Figure 1d–f. Figure 1d shows several graphene flakes, with a lateral size of several hundreds of nanometers. A thickness of seven graphene layers is observed at the flake edge as shown in the inset of Figure 1d. A high-resolution image is shown in Figure 1e, illustrating the crystalline structure and lattice fringes characteristic of graphene. The selected area electron diffraction (SAED) pattern, shown in Figure 1f, indicates polycrystalline

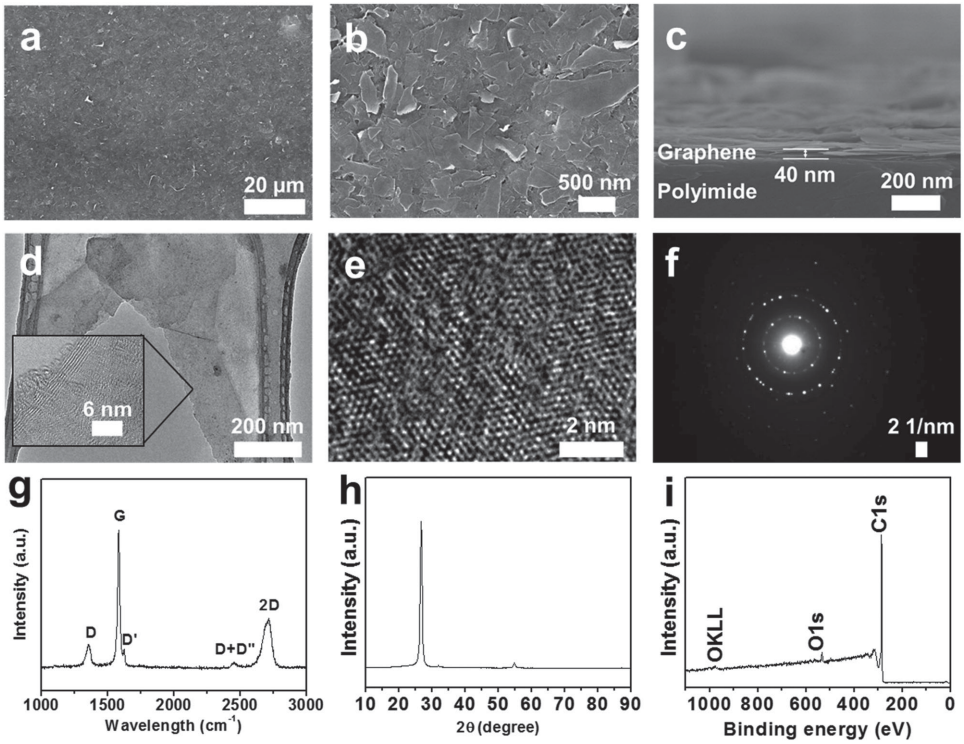


Figure 1. Structural and chemical characterization of graphene films. a–c) SEM images of graphene films on a polyimide substrate, including (a,b) top-view and (c) cross-sectional images. d,e) TEM images of graphene flakes at different magnifications, along with f) SAED pattern of graphene. g–i) Large-area characterization of graphene films including Raman spectroscopy, XRD, and XPS, respectively.

© 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

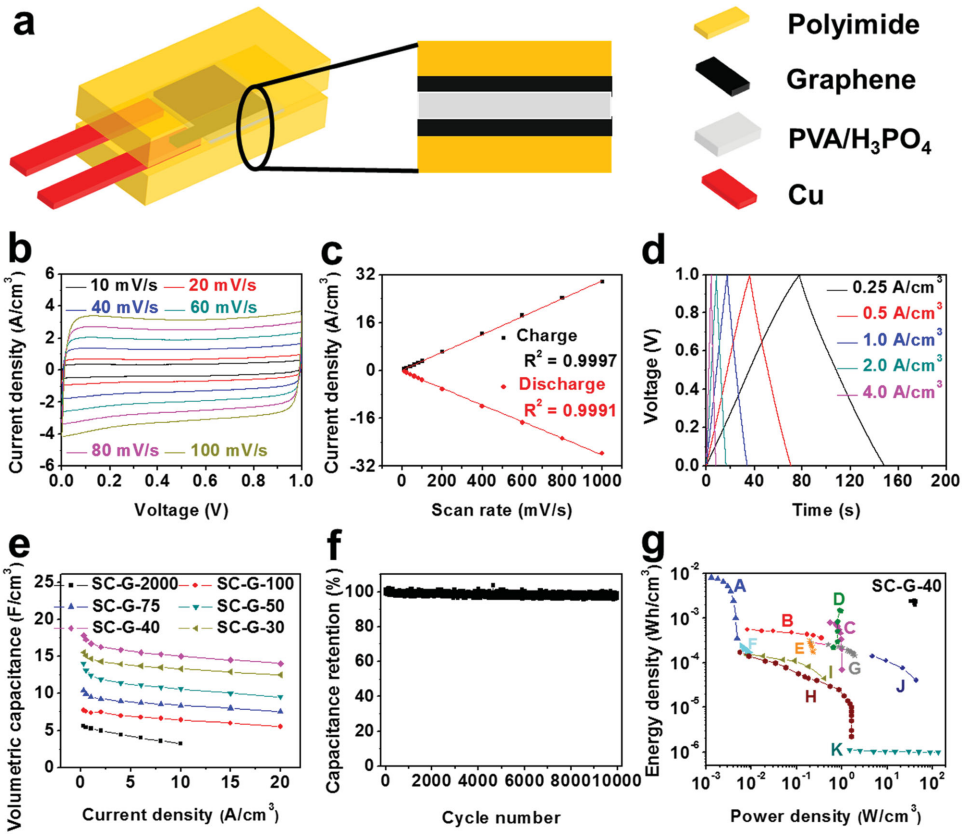


Figure 2. Electrochemical performance of sandwich-structured graphene supercapacitors. a) Schematic diagram of a sandwich-structured supercapacitor, including the 3D structure and cross-section (not to scale). b) CV curves for a device with 40 nm thick electrodes (SC-G-40) at different scan rates from 10 to 100 mV s^{-1}. c) The linear relationship between the current density (extracted from CV curves at 0.5 V for both charge and discharge processes) and scan rates for SC-G-40. d) Galvanostatic charge–discharge curves of SC-G-40 at different current densities ranging from 0.25 to 4.0 A cm^{-3}. e) Volumetric capacitance of devices with different electrode thicknesses, measured at different current densities. f) Cycling performance of SC-G-40 at a current density of 4.0 A cm^{-3}. g) Ragone plot of SC-G-40 and the comparison with commercially available energy storage devices and recently reported materials. A: Li thin-film battery 4 V/500 mAh,[24] B: Commercial SC 5.5 V/100 mF,[54] C: Commercial AC-SC 2.75 V/44 mF,[24] D: Co_9S_8/carbon cloth-Co_3O_4@RuO_2/carbon cloth (CSC-CRC),[45] E: Hydrogen-treated TiO_2@MnO_2//hydrogen-treated TiO_2 carbon shell (H-TiO_2@MnO_2//H-TiO_2@C),[47] F: Carbon nanofiber,[44] G: Laser-scribed graphene (LSG),[17] H: Methane (CH_4)-plasma treatment graphene oxide (MPG),[18] I: Electrochemically reduced graphene oxide (ErGO),[43] J: Hydrogenated single-crystal ZnO@amorphous ZnO-doped MnO_2 core–shell nanocables (HZM),[46] K: Al electrolytic capacitor 3 V/300 mF.[24]

features due to the stacking of multiple graphene flakes. Furthermore, large-area characterization of the graphene films including Raman spectroscopy (Figure 1g), X-ray diffraction (XRD) (Figure 1h), and X-ray photoelectron spectroscopy (XPS) (Figure 1i), indicate the high quality of the material.[26,32,34,38] Raman spectroscopy shows clear peaks for the D (1356 cm^{-1}), G (1581 cm^{-1}), D′ (1622 cm^{-1}), D+D″ (2455 cm^{-1}), and 2D (2712 cm^{-1}) bands.[35,39] The low Raman D-to-G peak ratio suggests a low density of defects, while the sharp XRD peak is correlated with the crystallinity of the few-layer graphene. The XPS spectrum further indicates a low oxygen content of the film and minimal C—O functionality, resulting from the use of pristine rather than oxidized graphene (Figure S4, Supporting Information).

The electrochemical performance of sandwich-structured graphene supercapacitors (**Figure 2**a) was first studied by cyclic voltammetry (CV) and galvanostatic charge–discharge experiments in a potential window from 0.0 to 1.0 V. Figure 2b and Figure S5 (Supporting Information) show the CV curves of SC-G-40 at different scan rates from 10 to 1000 mV s^{-1}. The CV curves exhibit a highly rectangular shape at low scan rate, which was well-maintained even at a high scan rate of 100 mV s^{-1}. The CV shape is distorted only at exceptionally high scan rates due to internal resistance, especially at a scan rate of 1000 mV s^{-1}.[40] The capacitive current increased with an increasing scan rate, with a strong linear correlation observed with R^2 = 0.9997 and 0.9991 for the charge and discharge processes, respectively (Figure 2c). The results demonstrate the desirable capacitive behavior of SC-G-40. Similar performance characteristics were observed in other SC-G-*X* devices, as shown in Figure S6 (Supporting Information). Figure 2d and Figure S7 (Supporting Information) show the galvanostatic charge–discharge curves of these devices with varying current densities. The nearly symmetric charge and discharge curves confirm the desirable capacitive behavior of these devices, consistent with the CV results.

The volumetric capacitance of these devices and their electrodes are calculated from the galvanostatic charge–discharge

 © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

curves, and plotted in Figure 2e and Figure S8a (Supporting Information), respectively. By these metrics, SC-G-40 yielded the best performance among this set of devices. Specifically, the volumetric capacitance reaches 17.8 F cm^{-3} at a current density of 0.25 A cm^{-3} for the full device and 71.2 F cm^{-3} at a current density of 0.5 A cm^{-3} for each individual electrode. With increasing current density, the device capacitance gradually decreases to 14.0 F cm^{-3} at a current density of 20 A cm^{-3}. This behavior represents excellent rate performance, with a capacitance retention of 79% as the current density increases by a factor of 80 from 0.25 to 20 A cm^{-3}. Other devices show a similar trend with varying current densities. The thickness of the electrodes affects the device performance, especially for SC-G-2000. The volumetric capacitance of this device was 5.6 F cm^{-3} at a current density of 0.25 A cm^{-3}. It decreased to 3.2 F cm^{-3} when the current density increased to 10 A cm^{-3}, with only 57% capacitance retained, likely due to limited ion diffusion in the electrodes.[7] This conclusion is further supported by the observation that devices with thinner electrodes exhibit higher volumetric capacitance (Figure S8b, Supporting Information). Electrodes thinner than 40 nm, however, show a decrease in volumetric capacitance, possibly due to limited electrical conductance in the plane of the film.[32] The graphene film exhibits a packing density of 1.12 g cm^{-3} (Figure S9, Supporting Information), higher than that of graphene films prepared by other methods with densities ranging from 0.05 to 0.75 g cm^{-3}.[41,42] The gravimetric capacitance is promising, as shown in Figure S10 (Supporting Information), with an electrode gravimetric capacitance of 63.6 F g^{-1} at a current density of 0.5 A cm^{-3} for SC-G-40. In addition, the higher packing density for a given gravimetric capacitance enables lower electrolyte usage, thereby limiting the overall device mass. The cycling stability of these sandwich-structure supercapacitors was also evaluated through extended galvanostatic charge–discharge experiments, as shown in Figure 2f and Figure S11 (Supporting Information). After 10 000 cycles of charge–discharge sweeps, no systematic decay in capacitance is observed for these devices, confirming their excellent cycling stability.

Figure 2g and Figure S12 (Supporting Information) show Ragone plots of the volumetric energy density and power density of graphene supercapacitors, as well as a comparison with commercially available energy storage devices and recently reported results for solid-state supercapacitors. Table S1 (Supporting Information) summarizes the performance of these six devices at different conditions. SC-G-40 yields the best performance, with an energy density of 2.47 mWh cm^{-3} and a power density of 40.3 W cm^{-3}. This energy density is much higher than typical commercial supercapacitors (2.75 V/44 mF and 5.5 V/100 mF), and even comparable to Li thin-film batteries (4 V/500 μAh), while the power density is more than ten times higher than commercial SCs and more than 1000 times higher than Li thin-film batteries. SC-G-40 also compares favorably to other reported carbon materials in solid-state devices such as electrochemically reduced graphene oxide (ErGO),[43] carbon nanofibers,[44] methane (CH_4)-plasma treatment graphene oxide (MPG),[18] and laser-scribed graphene (LSG),[17] as well as pseudocapacitive materials such as carbon cloth-Co_3O_4@RuO_2/carbon cloth (CSC-CRC),[45] hydrogenated single-crystal ZnO@amorphous ZnO-doped MnO_2 core–shell nanocables (HZM),[46] hydrogen-treated TiO_2@MnO_2//hydrogen-treated TiO_2 carbon shell (H-TiO_2@MnO_2//H-TiO_2-C),[47] and nanoporous gold-polypyrrole (NPG-PPy).[48] These competitive metrics, coupled with design simplicity and processing advantages, support the promise of our printable graphene electrodes for practical applications.

Graphene films cast from a pristine graphene ink show desirable performance in sandwich-structured SCs, as discussed above. A key advantage of this material is its broad process compatibility, particularly its versatility for different solution-phase patterning methods. For example, G/EC inks have been developed for inkjet printing, a prevalent additive manufacturing technology with numerous benefits including digital, noncontact, and low-waste patterning.[49,50] In an effort to exploit these advantages, microsupercapacitors with inkjet-printed interdigitated graphene electrodes were fabricated on flexible polyimide substrates, as illustrated in Figure S13 (Supporting Information). In particular, the graphene ink was inkjet printed onto polyimide to form a G/EC pattern containing 12 in-plane interdigitated electrodes (six per polarity). Following removal of the EC, PVA-H_3PO_4 was cast onto the surface of the graphene patterns to complete fabrication of the MSCs. The digital photograph in Figure S13 (Supporting Information) shows one complete device, labeled as MSC-G. The graphene patterns can be printed in scale, offering compelling prospects for practical utility (Figure S14, Supporting Information).

CV and galvanostatic charge–discharge experiments were carried out to evaluate the electrochemical performance of MSC-G (**Figure 3**a). The nearly rectangular shape of the CV curves (Figure 3b and Figure S15a (Supporting Information)), strong linear relationship between capacitive current and scan rates (Figure 3c), and symmetric triangular charge–discharge curves (Figure 3d and Figure S15b (Supporting Information)) demonstrate the desirable capacitive behavior of these devices. Based on the charge–discharge experiments, the volumetric capacitance of the devices was measured. As shown in Figure 3e, the volumetric capacitance of the device was 9.3 F cm^{-3} at a current density of 0.25 A cm^{-3}, and decreased to 5.0 F cm^{-3} when the current density increased to 50 A cm^{-3}. This capacitance retention of greater than 54% for a corresponding current increase by a factor of 200 indicates reasonable rate performance, especially for a MSC. Importantly, the volume used in this capacitance calculation includes the space between the electrodes, thereby reflecting both the intrinsic performance of the graphene electrode and the fabrication tolerance and resolution; as such, it is expected that this capacitance is reduced from the sandwich-structured SCs (Figure S16, Supporting Information).

The cycling stability of MSC-G was evaluated by extended galvanostatic charge–discharge measurements. As shown in Figure 3f, minimal capacitance decay is observed after 10 000 cycles, indicating the excellent operational stability of the device. Volumetric energy and power density of MSC-G were also evaluated (1.29 mWh cm^{-3} and 278 W cm^{-3}, respectively), and are shown in Figure 3g along with the corresponding data for recent literature reports of carbon-based electrodes, indicating excellent performance for the G/EC MSCs in this broader context.[31,51–53] MSC fabrication commonly uses expensive lithography for electrode patterning, the addition of conductive agents or separate current collectors, and often requires

© 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

16146840, 2016, 20, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

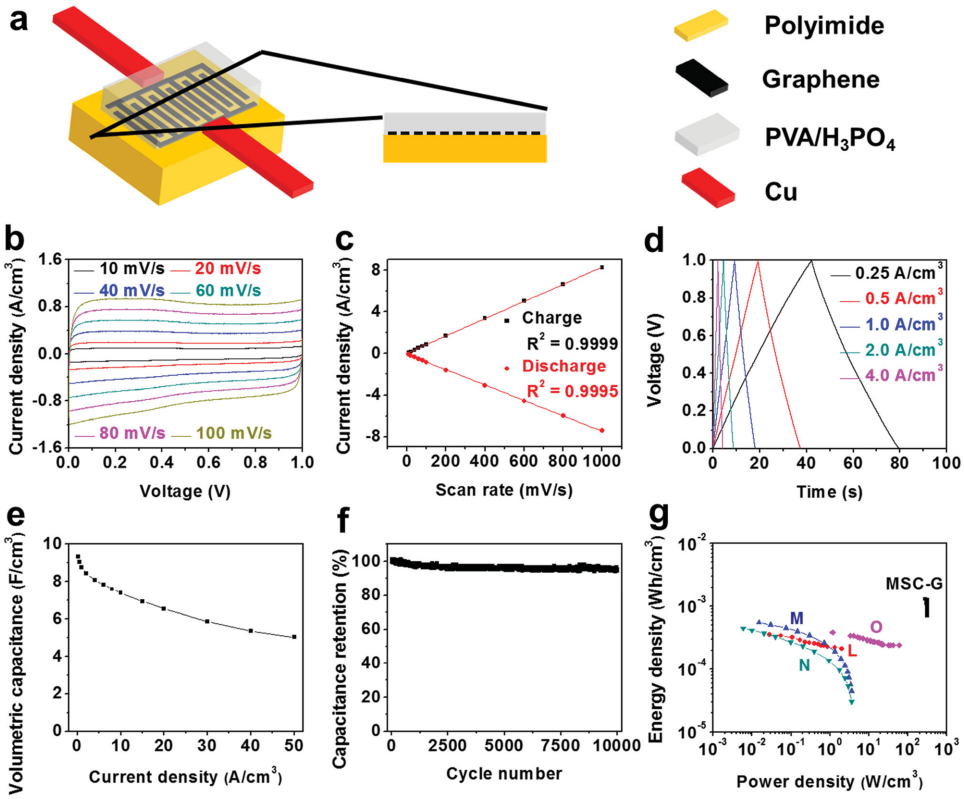


Figure 3. Electrochemical performance of inkjet-printed graphene microsupercapacitors. a) Schematic diagram of a microsupercapacitor with interdigitated structure, including both a 3D representation and 2D cross-section (not to scale). b) CV curves of MSC-G at different scan rates from 10 to 100 mV s^{-1}. c) The linear relationship between the current density (extracted from CV curves at 0.5 V for both charge and discharge processes) and scan rates for MSC-G. d) Galvanostatic charge–discharge curves of MSC-G at different current densities ranging from 0.25 to 4.0 A cm^{-3}. e) Volumetric capacitance of MSC-G at different current densities. f) Cycling performance of MSC-G at a current density of 4.0 A cm^{-3}. g) Ragone plot of MSC-G and recent literature reports. L: Carbon fiber tows (CFTs),[51] M: Boron-doped laser-induced graphene (B-LIG),[52] N: Laser-induced graphene (LIG),[31] O: Laser-scribed graphene (LSG).[53]

high temperature and multistep synthetic processes.[4] In this work, the fabrication of the electrodes is achieved through inkjet printing of graphene ink and subsequent EC removal by modest annealing in ambient atmosphere, representing a key processing advantage over alternative methods for MSC fabrication.

The ability to fabricate graphene MSCs in a shared processing platform with printed electronics motivates their application in powering flexible, portable devices. Toward that end, the durability of these devices under mechanical deformation is critical for reliable long-term operation. The device design presented here offers several advantages for mechanical durability. Notably, films of G/EC have been demonstrated to exhibit excellent resilience to extreme bending and even discrete folding, with minimal degradation in electrical performance.[32,37] In addition, by employing the highly conductive G/EC as both the current collector and electrode, the number of interfaces in the device is reduced. Materials and interfaces introduced with traditional metal current collectors can present mechanically weak points in terms of electrode fracture and delamination.[4,18] To evaluate these potential advantages, the performance of SC-G-40 and MSC-G was studied under bending mechanical strain. The nearly invariant capacitance measured at different strain states indicates the stable operation of the devices under this mechanical deformation (**Figure 4**a and Figure S17 (Supporting Information)). Moreover, when flexed repeatedly to a radius of curvature of 2.5 mm, corresponding to bending strains of 5.1% and 2.5% for the SC and MSC, respectively, these devices show desirable mechanical tolerance with no systematic capacitance decay over 5000 bending cycles, as shown in Figure 4b. This result also indicates strong adhesion between the graphene films and the polyimide substrate, because delamination and interfacial fracture is a common failure mechanism under bending stresses.

In addition to bending tolerance, devices for portable energy storage applications require sufficient robustness to provide reliable operation under abrupt mechanical impact. To demonstrate the suitability of the design presented here, electrochemical performance was evaluated following repeated impact, applied with a hammer swinging with ≈2 J of energy (Figure S18, Supporting Information). As shown in Figure 4c, no systematic change in performance of a graphene SC is observed over 100 cycles of repeated impact. The combination of bending tolerance and impact resistance reinforces the suitability of these devices for flexible and portable energy storage applications. In addition, to meet the specific energy

 © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

ADVANCED ENERGY MATERIALS
www.advenergymat.de

www.MaterialsViews.com

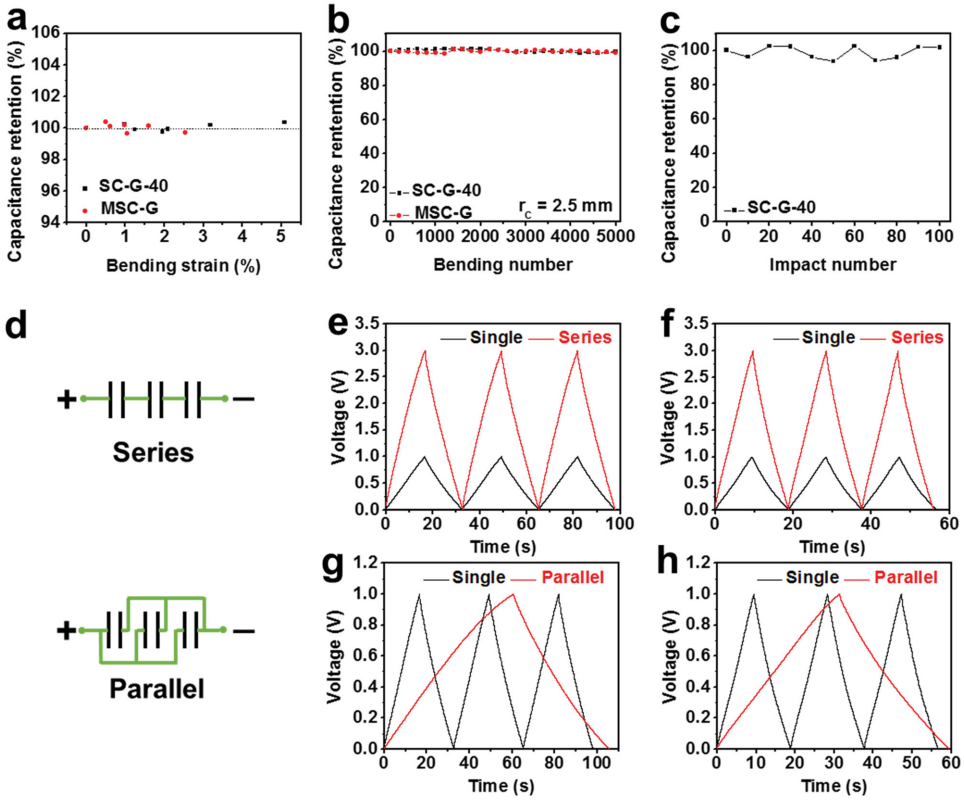


Figure 4. Flexibility and impact testing, as well as assembly of multiple devices in parallel and series configurations for SC-G-40 and MSC-G. a) Capacitance retention of SC-G-40 and MSC-G under different bending strains at a current density of 1.0 A cm^{-3}. b) Capacitance retention of SC-G-40 and MSC-G at different bending cycles to a radius of curvature of 2.5 mm. c) Capacitance retention of SC-G-40 following repeated impact testing. d) Schematic circuit diagram of three single devices connected in series and parallel configurations. e,f) Galvanostatic charge–discharge curves of three SC-G-40 and MSC-G devices, respectively, connected in series, and comparison with a single device at the same current (normalized to be 1.0 mA cm^{-2} for a single device). g,h) Galvanostatic charge–discharge curves of three SC-G-40 and MSC-G devices, respectively, connected in parallel, and comparison with a single device at the same current (normalized to be 1.0 mA cm^{-2} for a single device).

and power needs for practical applications, multiple SC-G-40 and MSC-G can be assembled in series or parallel configurations (Figure 4d). When three devices are connected in series, it results in an increased voltage window by a factor of 3, with a similar discharge time for a given current (Figure 4e,f). Compared with a single device, the discharge time of three devices connected in parallel increases by more than three times compared to that of a single device at the same current (Figure 4g,h), indicating that these devices can be combined in series or parallel to tailor the voltage or working time.

In conclusion, we have demonstrated a route for the fabrication of all-solid-state flexible SCs and MSCs using a solution-processed pristine graphene ink. The graphene electrodes demonstrate superlative electrical, electrochemical, and mechanical properties, along with desirable process compatibility. Specifically, the solid-state flexible devices exhibit high volumetric capacitance, promising energy and power densities, and excellent cycling stability and mechanical durability. These results provide a compelling platform to simplify energy storage device fabrication processes, with promise for scalable manufacturing, digital device design, and direct integration with printed electronic systems. Overall, the design strategy discussed here offers a new avenue for producing robust, high-performance, all-solid-state flexible energy storage devices in a scalable, straightforward, and versatile process.

Experimental Section

Synthesis of G/EC: Graphene was produced by high shear mixing of graphite in a solution of ethyl cellulose and ethanol. Ethyl cellulose (EC, Sigma-Aldrich, 4 cP grade measured in 80:20 toluene:ethanol at 5 wt%, 48% ethoxy) was dissolved in ethanol (Koptec, 200 proof) at a concentration of 10 mg mL^{-1}. Flake graphite (Asbury Graphite Mills, Grade 3061) was added at a concentration of 200 mg mL^{-1}. This mixture (≈1 L total volume) was mixed with a high shear mixer (Silverson L5M-A) for 2 h at 10 230 rpm in an ice water bath. Unexfoliated graphite flakes were removed by centrifugation (Beckman Coulter Avanti J-26 XPI centrifuge) at 7500 rpm for 5 min and then 4000 rpm for 105 min. The supernatant was collected and flocculated by mixing with an aqueous solution of NaCl (Sigma-Aldrich, 0.04 g mL^{-1}) in a 16:9 weight ratio. Centrifugation at 7500 rpm for 6 min was performed to collect the flocculated G/EC solid, and the ethanol and excess EC was discarded. Finally, the G/EC solid was washed with deionized water to remove residual salt and dried.

Fabrication of the Sandwich-Structured SCs and MSCs: Graphene supercapacitors were prepared by spin-coating and blade-coating

© 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

processes. For blade-coating thick graphene films, G/EC powder was dispersed in ethanol and terpineol by bath sonication. Excess ethanol was removed by heating to yield a nominal ink composition of 20% w/v G/EC in terpineol. This paste was doctor-bladed onto 125 μm thick polyimide substrates to form uniform films. Following EC removal by thermal annealing, the graphene film thickness was ≈2000 nm. For spin-coating thin graphene films, 0.2 mL of blade-coating ink was diluted in 0.5 mL ethanol. This spin-coating ink was cast into films on polyimide with a spin speed of 1000–5000 rpm. Films were thermally annealed at 350 °C for 4 h to remove the EC prior to electrolyte deposition. Polymeric gel electrolyte of PVA/H_3PO_4 was prepared by stirring 9 mL of deionized water, 1.0 mL of phosphoric acid (85%, Sigma-Aldrich), and 1.0 g of PVA (M_w = 50 000, Aldrich) at 80 °C. The electrolyte was deposited on the active area of the devices, and was dried under ambient conditions for 4 h. The all-solid-state SCs were obtained after drying in a vacuum desiccator overnight for further solidification of the electrolyte.

Graphene microsupercapacitors were prepared by inkjet printing. G/EC powder was dispersed at a concentration of 30 mg mL^{-1} in a solvent system containing 80:15:5 v/v cyclohexanone, terpineol, and di(ethylene glycol) methyl ether (Sigma-Aldrich). Following bath sonication to disperse the powder, the ink was passed through a 3.1 μm glass fiber syringe filter to prevent nozzle clogging and promote stable printing. This graphene ink was stable for a period of at least six months. A Ceradrop X-Serie inkjet printer equipped with a 10 pL Dimatix cartridge (DMC-11610) was used to print the ink using a custom waveform to form the interdigitated electrode structure of the MSCs. All electrodes were printed on 125 μm thick polyimide and annealed at 350 °C for 4 h prior to electrolyte deposition. The same process for the electrolyte as discussed for SCs was used for MSCs.

Electrochemical Characterization: The electrochemical performance of SCs and MSCs were characterized by CV and galvanostatic charge–discharge experiments (CHI 760D). The volumetric capacitance (C_V) of electrode materials were calculated based on galvanostatic charge–discharge curves according to Equation (1)

$$C_V = 4I/\left(V_{Device} \times \left(dV/dt\right)\right) \quad (1)$$

where I was the current applied, V_{Device} was the total volume of the device, and dV/dt was the slope of the discharge curve.

The volumetric capacitance ($C_{Device,V}$) of SC and MSC devices were calculated by using Equation (2)

$$C_{Device,V} = C_V/4 \quad (2)$$

The volumetric energy density ($E_{Device,V}$) of SCs and MSCs were calculated by using Equation (3)

$$E_{Device,V} = C_{Device,V}V^2/(2 \times 3600) \quad (3)$$

where V was the applied voltage.

The volumetric power density ($P_{Device,V}$) of SCs and MSCs were calculated by using Equation (4)

$$P_{Device,V} = V^2/4R_{ESR}V_{Device} \quad (4)$$

where R_{ESR} was the internal resistance of the device, which was estimated from the voltage drop (V_{Drop}) at the beginning of the discharge at an applied current (I) according to Equation (5)

$$R_{ESR} = V_{Drop}/2I \quad (5)$$

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

L.L. and E.B.S. contributed equally to this work. The SC and MSC fabrication and testing work was supported by the Center for Electrochemical Energy Science (CEES), an Energy Frontier Research Center (EFRC) funded by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences (Award No. DE-AC02–06CH11357). The graphene ink preparation was supported by the Air Force Research Laboratory under agreement number FA8650-15-2-5518. E.B.S. received further support from the Department of Defense (DoD) through the National Defense Science and Engineering Graduate (NDSEG) Fellowship Program, as well as the Ryan Fellowship administered by the Northwestern University International Institute for Nanotechnology. Characterization made use of the NUANCE facility at Northwestern University, which received support from the NSF-MRSEC (DMR-1121262), Keck Foundation, and the State of Illinois. The U. S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright notation thereon. The views and conclusions contained herein were those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the sponsors.

Received: May 1, 2016
Revised: June 20, 2016
Published online: July 21, 2016

[1] X. Wang, X. Lu, B. Liu, D. Chen, Y. Tong, G. Shen, *Adv. Mater.* **2014**, *26*, 4763.
[2] A. Nathan, A. Ahnood, M. T. Cole, L. Sungsik, Y. Suzuki, P. Hiralal, F. Bonaccorso, T. Hasan, L. Garcia-Gancedo, A. Dyadyusha, S. Haque, P. Andrew, S. Hofmann, J. Moultrie, C. Daping, A. J. Flewitt, A. C. Ferrari, M. J. Kelly, J. Robertson, G. Amaratunga, W. I. Milne, *Proc. IEEE* **2012**, *100*, 1486.
[3] L. L. Zhang, X. S. Zhao, *Chem. Soc. Rev.* **2009**, *38*, 2520.
[4] M. Beidaghi, Y. Gogotsi, *Energy Environ. Sci.* **2014**, *7*, 867.
[5] G. Wang, L. Zhang, J. Zhang, *Chem. Soc. Rev.* **2012**, *41*, 797.
[6] P. Simon, Y. Gogotsi, *Nat. Mater.* **2008**, *7*, 845.
[7] Y. Xu, Z. Lin, X. Zhong, X. Huang, N. O. Weiss, Y. Huang, X. Duan, *Nat. Commun.* **2014**, *5*, 4554.
[8] Y. He, W. Chen, C. Gao, J. Zhou, X. Li, E. Xie, *Nanoscale* **2013**, *5*, 8799.
[9] Y. Huang, J. Liang, Y. Chen, *Small* **2012**, *8*, 1805.
[10] X. Wang, G. Shi, *Energy Environ. Sci.* **2015**, *8*, 790.
[11] L. L. Zhang, R. Zhou, X. S. Zhao, *J. Mater. Chem.* **2010**, *20*, 5983.
[12] W. S. Hummers, R. E. Offeman, *J. Am. Chem. Soc.* **1958**, *80*, 1339.
[13] N. I. Kovtyukhova, P. J. Ollivier, B. R. Martin, T. E. Mallouk, S. A. Chizhik, E. V. Buzaneva, A. D. Gorchinskiy, *Chem. Mater.* **1999**, *11*, 771.
[14] D. C. Marcano, D. V. Kosynkin, J. M. Berlin, A. Sinitskii, Z. Sun, A. Slesarev, L. B. Alemany, W. Lu, J. M. Tour, *ACS Nano* **2010**, *4*, 4806.
[15] D. Li, M. B. Muller, S. Gilje, R. B. Kaner, G. G. Wallace, *Nat. Nanotechnol.* **2008**, *3*, 101.
[16] W. Gao, L. B. Alemany, L. Ci, P. M. Ajayan, *Nat. Chem.* **2009**, *1*, 403.
[17] M. F. El-Kady, V. Strong, S. Dubin, R. B. Kaner, *Science* **2012**, *335*, 1326.
[18] Z. S. Wu, K. Parvez, X. Feng, K. Müllen, *Nat. Commun.* **2013**, *4*, 2487.
[19] M. J. McAllister, J.-L. Li, D. H. Adamson, H. C. Schniepp, A. A. Abdala, J. Liu, M. Herrera-Alonso, D. L. Milius, R. Car, R. K. Prud'homme, I. A. Aksay, *Chem. Mater.* **2007**, *19*, 4396.
[20] X. Fan, W. Peng, Y. Li, X. Li, S. Wang, G. Zhang, F. Zhang, *Adv. Mater.* **2008**, *20*, 4490.

 © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[21] D. Luo, G. Zhang, J. Liu, X. Sun, *J. Phys. Chem. C* **2011**, *115*, 11327.
[22] A. M. Dimiev, L. B. Alemany, J. M. Tour, *ACS Nano* **2013**, *7*, 576.
[23] Z. L. Wang, *Nano Today* **2010**, *5*, 512.
[24] D. Pech, M. Brunet, H. Durou, P. Huang, V. Mochalin, Y. Gogotsi, P.-L. Taberna, P. Simon, *Nat. Nanotechnol.* **2010**, *5*, 651.
[25] X. Li, W. Cai, J. An, S. Kim, J. Nah, D. Yang, R. Piner, A. Velamakanni, I. Jung, E. Tutuc, S. K. Banerjee, L. Colombo, R. S. Ruoff, *Science* **2009**, *324*, 1312.
[26] Z. Chen, W. Ren, L. Gao, B. Liu, S. Pei, H.-M. Cheng, *Nat. Mater.* **2011**, *10*, 424.
[27] Z. Sun, Z. Yan, J. Yao, E. Beitler, Y. Zhu, J. M. Tour, *Nature* **2010**, *468*, 549.
[28] S. Zhou, J. Xu, Y. Xiao, N. Zhao, C.-P. Wong, *Nano Energy* **2015**, *13*, 458.
[29] K. Liu, Y.-M. Chen, G. M. Policastro, M. L. Becker, Y. Zhu, *ACS Nano* **2015**, *9*, 6041.
[30] J. Lin, Z. Peng, Y. Liu, F. Ruiz-Zepeda, R. Ye, E. L. G. Samuel, M. J. Yacaman, B. I. Yakobson, J. M. Tour, *Nat. Commun.* **2014**, *5*, 5714.
[31] Z. Peng, J. Lin, R. Ye, E. L. G. Samuel, J. M. Tour, *ACS Appl. Mater. Interfaces* **2015**, *7*, 3414.
[32] E. B. Secor, P. L. Prabhumirashi, K. Puntambekar, M. L. Geier, M. C. Hersam, *J. Phys. Chem. Lett.* **2013**, *4*, 1347.
[33] E. B. Secor, S. Lim, H. Zhang, C. D. Frisbie, L. F. Francis, M. C. Hersam, *Adv. Mater.* **2014**, *26*, 4533.
[34] Y. T. Liang, M. C. Hersam, *J. Am. Chem. Soc.* **2010**, *132*, 17661.
[35] E. B. Secor, B. Y. Ahn, T. Z. Gao, J. A. Lewis, M. C. Hersam, *Adv. Mater.* **2015**, *27*, 6683.
[36] E. B. Secor, M. C. Hersam, *J. Phys. Chem. Lett.* **2015**, *6*, 620.
[37] W. J. Hyun, E. B. Secor, G. A. Rojas, M. C. Hersam, L. F. Francis, C. D. Frisbie, *Adv. Mater.* **2015**, *27*, 7058.
[38] Y. Zhu, L. Li, C. Zhang, G. Casillas, Z. Sun, Z. Yan, G. Ruan, Z. Peng, A.-R. O. Raji, C. Kittrell, R. H. Hauge, J. M. Tour, *Nat. Commun.* **2012**, *3*, 1225.
[39] A. C. Ferrari, D. M. Basko, *Nat. Nanotechnol.* **2013**, *8*, 235.
[40] B.-J. Yoon, S.-H. Jeong, K.-H. Lee, H. Seok Kim, C. Gyung Park, J. Hun Han, *Chem. Phys. Lett.* **2004**, *388*, 170.
[41] P. Simon, Y. Gogotsi, *Acc. Chem. Res.* **2013**, *46*, 1094.
[42] S. Murali, N. Quarles, L. L. Zhang, J. R. Potts, Z. Tan, Y. Lu, Y. Zhu, R. S. Ruoff, *Nano Energy* **2013**, *2*, 764.
[43] B. Xie, C. Yang, Z. Zhang, P. Zou, Z. Lin, G. Shi, Q. Yang, F. Kang, C.-P. Wong, *ACS Nano* **2015**, *9*, 5636.
[44] Y. Sun, R. B. Sills, X. Hu, Z. W. Seh, X. Xiao, H. Xu, W. Luo, H. Jin, Y. Xin, T. Li, Z. Zhang, J. Zhou, W. Cai, Y. Huang, Y. Cui, *Nano Lett.* **2015**, *15*, 3899.
[45] J. Xu, Q. Wang, X. Wang, Q. Xiang, B. Liang, D. Chen, G. Shen, *ACS Nano* **2013**, *7*, 5453.
[46] P. Yang, X. Xiao, Y. Li, Y. Ding, P. Qiang, X. Tan, W. Mai, Z. Lin, W. Wu, T. Li, H. Jin, P. Liu, J. Zhou, C. P. Wong, Z. L. Wang, *ACS Nano* **2013**, *7*, 2617.
[47] X. Lu, M. Yu, G. Wang, T. Zhai, S. Xie, Y. Ling, Y. Tong, Y. Li, *Adv. Mater.* **2013**, *25*, 267.
[48] F. Meng, Y. Ding, *Adv. Mater.* **2011**, *23*, 4098.
[49] Y. Aleeva, B. Pignataro, *J. Mater. Chem. C* **2014**, *2*, 6436.
[50] B. Derby, *Annu. Rev. Mater. Res.* **2010**, *40*, 395.
[51] D. Yu, S. Zhai, W. Jiang, K. Goh, L. Wei, X. Chen, R. Jiang, Y. Chen, *Adv. Mater.* **2015**, *27*, 4895.
[52] Z. Peng, R. Ye, J. A. Mann, D. Zakhidov, Y. Li, P. R. Smalley, J. Lin, J. M. Tour, *ACS Nano* **2015**, *9*, 5868.
[53] M. F. El-Kady, R. B. Kaner, *Nat. Commun.* **2013**, *4*, 1475.
[54] W. Zuo, C. Wang, Y. Li, J. Liu, *Sci. Rep.* **2015**, *5*, 7780.

© 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16146840, 2016, 20, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.201600909 by Yichao Zhao - Massachusetts Institute of Technolo , Wiley Online Library on [29/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License